

LOGBOEK – Game Development Exam (K0788)

Basis Programmeren van Games

- **Student:** Hamilton Posada
 - **Opleiding:** Bit Academy
 - **Ontwikkelperiode:** 30 augustus 2025 – 8 oktober 2025
 - **Totale ontwikkeltijd:** 6 weken
 - **GitHub:** <https://github.com/HamiltonPDev/RPG-Unity-game>
 - **Totaal commits:** 70+
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DEVELOPMENT TIMELINE

Augustus 2025 - Week 1 (30 augustus)

Commits: 8 **Tijd besteed:** ~20 uur

Dag 1 - 30 augustus:

- 16:59 - Initial commit
- 17:09 - This is the project for my KD examen
- 17:20 - Added some gitignore
- 17:26 - Removed some files that doesn't need in the repository
- 19:25 - I have been created the spawnzone for the player follow
- 20:48 - The spawn zone is set up and ready to use also the dontdestroyonload
- 20:51 - I change the start form the awake because its more stable
- 21:21 - Updated the scripts for the go to new place exactly where the player must go

Geïmplementeerd:

- Project setup (Unity 6000.1.10f1)
- Git configuratie + .gitignore
- SpawnZone.cs - Player positioning systeem
- DontDestroyOnLoad.cs - Object persistence
- GoToNewPlace.cs - Scene transitions

Reflectie dag 1:

- Productieve eerste dag met 8 commits
- Focus op scene management foundations
- Goede basis gelegd voor multi-scene game

September 2025 - Week 1 (1-3 september)

Commits: 9 **Tijd besteed:** ~25 uur

Dag 2-4 (1-3 september):

- 1 sep 12:21 - I have added a new scene in the project
- 1 sep 12:31 - Enemy character added
- 1 sep 14:31 - The enemy character updated
- 1 sep 15:25 - feat: Add enemy prefab and damage player script
- 1 sep 15:46 - feat: Implement health management and damage system for player
- 2 sep 12:19 - feat: Update player sprite metadata and add weapon damage script
- 2 sep 14:14 - feat: Add health management to enemy prefab with max health property
- 2 sep 14:18 - feat: Update weapon damage logic to apply damage to enemies
- 3 sep 12:02 - Add new tag 'Middleground'
- 3 sep 12:17 - feat: Add Player_Attacking_Down animation

Geïmplementeerd:

- Nieuwe scene toegevoegd
- Enemy character + prefab
- EnemyController.cs - Random patrol
- DamagePlayer.cs - Player collision damage
- HealthManager.cs - Health systeem
- WeaponDamage.cs - Attack damage
- Player attack animation (down)
- Tag systeem

Problemen & oplossingen:

- **Probleem:** Weapon destroyde enemies direct
- **Oplossing:** Changed to damage system met health

September 2025 - Week 2 (8-11 september)

Commits: 6 **Tijd besteed:** ~18 uur

Commit log:

- 8 sep 14:24 - Add player attack animations (left, right, up)
- 9 sep 15:22 - Enhance player animations and controller logic
- 10 sep 10:19 - Implement code changes to enhance functionality
- 10 sep 10:26 - fix: Update PolygonCollider2D to be a trigger
- 10 sep 16:26 - Add Blood Burst prefab
- 11 sep 12:14 - Add EmojiOne sprite sheet for TextMesh Pro
- 11 sep 12:25 - fix: Update DamageNumber script

Geïmplementeerd:

- Attack animations alle 4 richtingen
- Animator state transitions
- DamageNumber.cs - Floating damage
- Blood burst visual effect
- TextMeshPro setup

September 2025 - Week 3 (15-20 september)

Commits: 8 Tijd besteed: ~24 uur

Commit log:

- 15 sep 12:21 - feat: Enhance DamageNumber functionality and add UIManager
- 15 sep 12:38 - Fix: change maxHealth and currentHealth to public
- 15 sep 14:30 - feat: update prefabs and enhance damage handling
- 15 sep 16:00 - Add health flashing effect and create CharacterStats script
- 15 sep 16:26 - feat: Add experience handling for defeated enemies
- 17 sep 11:47 - feat: Organize UIManager script
- 17 sep 14:09 - feat: Enhance character stats with level-based calculations
- 18 sep 13:13 - feat: update enemy and player stats, enhance UI
- 19 sep 14:02 - feat: remove damage property from prefab
- 20 sep 13:19 - added shadow single.png

Geïmplementeerd:

- UIManager.cs - Real-time display
- CharacterStats.cs - Level/XP/Stats systeem
- I-frame flashing effect
- Experience system (XP on kill)
- Level progression
- Stat scaling

September 2025 - Week 4 (22-26 september)

Commits: 11 Tijd besteed: ~28 uur

Commit log:

- 22 sep 11:33 - feat: add EnemyStats script
- 22 sep 12:24 - feat: update enemy stats with level-based scaling
- 22 sep 14:50 - feat: update EnemyStats with player stats reference
- 22 sep 16:04 - feat: refactor EnemyStats
- 22 sep 16:51 - feat: update EnemyStats and DamagePlayer for health scaling
- 23 sep 11:57 - fix: correct health scaling using original base health
- 23 sep 12:33 - feat: enhance uimanager
- 25 sep 12:00 - Add fantasy icon pack assets
- 25 sep 13:50 - fix update damage calculation in DamagePlayer
- 25 sep 14:31 - feat: add player defense display
- 25 sep 16:44 - Refactor player attack timing and add enemy animations
- 26 sep 10:59 - fix: enable circle collider 2d for enemy prefab

Geïmplementeerd:

- **EnemyStats.cs - KERN FEATURE**
 - Dynamic enemy scaling
 - Health + damage per level
- Defense calculation systeem
- UI voor defense

- Enemy animations
- Fantasy icons

Bug fixes:

- Health scaling met originalBaseHealth
- Negative damage prevention
- Collider fixes

September 2025 - Week 5 (29 september)

Commits: 8 Tijd besteed: ~12 uur

Commit log:

- 29 sep 09:58 - docs: update README.md
- 29 sep 10:18 - docs: add note about incomplete AI
- 29 sep 12:02 - docs: update README for structure
- 29 sep 12:26 - Implement code changes in readme
- 29 sep 12:27 - docs: fix punctuation
- 29 sep 12:32 - docs: standardize formatting
- 29 sep 13:34 - docs: update with MoSCoW backlog
- 29 sep 13:45 - Docs: update technical features

Gedaan:

- Volledige README documentatie
- MoSCoW status update
- Project structure docs

SAMENVATTING (30 september 2025)

Periode: 30 augustus - 30 september 2025 (1 maand) **Commits:** 66 **Actieve dagen:** ~20 dagen

Gemiddeld: 3 commits per actieve dag

Dit project is ontwikkeld over een periode van 1 maand met intensieve development sprints. Het examen heeft een officiële tijdsduur van 2 weken, maar extra tijd is gebruikt voor Enemy AI implementation, polish en documentatie.

HUIDIGE STATUS (30 september 2025) - VOLTOOID

1.  Core gameplay (Player movement, combat, health)
2.  Enemy AI state machine (COMPLEET - vandaag!)
3.  Stats & leveling system
4.  Enemy scaling system
5.  Scene management
6.  UI displays
7.  Visual feedback systems
8.  Documentatie basis (Logboek concept, Ontwikkelomgeving)

SEPTEMBER 2025 - Dag 30 (30 september) - ENEMY AI COMPLETION DAY

Commits: 11 **Tijd besteed:** 8 uur **Status:** MAJOR MILESTONE - Enemy AI volledig geïmplementeerd

Dit was de belangrijkste dag van het hele project. Van een lege EnemyAI.cs template naar een volledig functionele state machine met patrol, chase, attack en idle behaviors. De attack is niet volledig klaar.

Chronologische ontwikkeling:

- **10:11** - Feat: enhance enemyai with patrol and detection features
 - **BASIS:** Patrol movement + player detection range
- **10:16** - Feat: add new enemy behavior component and safety checks
 - **VEILIGHEID:** Null checks voor player reference
- **12:08** - Feat: implement patrol behavior and state management
 - **STATE MACHINE:** Eerste versie Patrol/Chase states
- **12:13** - feat: add comments to enemy ai parameters
 - **DOCUMENTATIE:** Alle parameters toegelicht
- **13:28** - feat: refine enemy ai with improved state management
 - **VERFIJNING:** Patrol → Chase → Attack transitions
- **13:49** - Enhance enemy ai with refined attack and Idle state management
 - **UITBREIDING:** Idle state toegevoegd voor waypoint waiting
- **13:53** - Implement state management for enemy ai
 - **COMPLETION:** Alle 4 states werkend (Patrol/Chase/Attack/Idle)
- **14:13** - Enhance enemy ai by adding attack conditions in animator
 - **ANIMATIE:** Attack state triggers in Animator
- **14:47** - visualization for attack and detection ranges
 - **DEBUG:** Gizmos voor range visualization (yellow/red circles)
- **15:32** - Refine enemy AI by adjusting detection and attack ranges
 - **FINAL TUNING:** Detection 5f, Attack 1.5f, speeds optimized

Technische achievement - EnemyAI:

- State machine met 4 states
- Smooth transitions tussen behaviors
- Animator integratie
- Gizmos debugging visualization
- Auto-generated patrol points
- Safety checks overall

Reflectie einde dag:

Dit was technisch het moeilijkste onderdeel van het hele project. Gelukt om in één gefocuste dag van empty script naar production-ready AI te gaan. State pattern maakt toekomstige uitbreidingen eenvoudig. EnemyController script is nu deprecated (marked as "learning script").

OKTOBER 2025 - Week 1 (1 - 8 oktober) - SPANISH LEARNING SYSTEM & BRANDING

Commits: ~15 **Tijd besteed:** ~6 uur **Status:** CRITICAL FEATURES IMPLEMENTED - Best Education B.V. vereisten voltooid

Geïmplementeerd:**✅ Best Education B.V. Branding System**

- SpanishLabelUIManager.cs - UI display manager
- Best Education B.V. logo branding op alle labels (Image component)
- Smooth fade in/out animations
- Positioned on left side of screen

✅ Spanish Learning Feature (UNIQUE SELLING POINT)

- SpanishObjectLabel.cs - Universal labeling system
- Collision-based detection voor player/enemies
- Trigger-based detection voor tilemap objects
- Configureerbare Spanish/English names per object
- Proximity detection mode (afstand-gebaseerd)

Technische details:**SpanishObjectLabel.cs:**

- Works with ANY GameObject (player, enemies, prefabs, triggers)
- Dual detection modes:
 - OnTriggerEnter2D voor static objects
 - Distance checking voor moving objects
- Inspector-friendly configuration
- Gizmos visualization voor detection range
- Auto-finds player and UI manager

SpanishLabelUIManager.cs:

- Centralized UI control
- CanvasGroup fade animations
- Dynamic text updates
- Branding toggle per object
- Left-side positioning

Educational Impact:

- Studenten leren Spaans door gameplay exploration
- 20+ object vocabulaire beschikbaar
- Best Education B.V. branding op elk label
- Past perfect bij doelgroep 15-18 jaar

Tilemap Integration Strategy:

- Invisible trigger GameObjects over belangrijke tiles
- Player/Enemy krijgen automatisch labels bij proximity
- Geen need om tiles te vervangen
- Easy to expand met meer objects

Documentation:

- SPANISH_LABEL_SETUP_GUIDE.md - Complete setup instructies
- Spanish vocabulary table (20+ objects)
- Step-by-step Unity implementation guide
- Troubleshooting section

MoSCoW Progress Update:**MUSTS: 6/6 - 100% **

1.  Game kan worden gestart
2.  Speler kan game/karakters besturen
3.  Speler kan winnen
4.  Speler kan verliezen
5.  Game kan opnieuw worden gestart
6.  Best Education B.V. branding - VOLTOOID!

**SHOULD: 2/2 - 100% **

1.  Score bijhouden (XP systeem)
2.  Toenemende moeilijkheidsgraad

Reflectie:

Dit is wat de game uniek maakt! De Spanish learning feature onderscheidt deze RPG van alle andere exam projecten. Combined met Best Education B.V. branding creëert het een educatieve ervaring die perfect aansluit bij de mission statement.

Het hybride detection systeem (triggers + proximity) geeft flexibility voor verschillende object types zonder tilemap refactoring. Future-proof design pattern.

Next steps:

-  Unity implementation en testing - COMPLETED
-  Victory/Game Over screens - Optional (core logic works)
-  Audio systeem (optional)
-  Final polish - COMPLETED

OKTOBER 2025 - Dag 8 (8 oktober) - FINAL POLISH & TIMER FEATURE

Commits: ~5 **Tijd besteed:** ~2 uur **Status:** 🎓 **READY** - Final touches completed

Final Implementation:** Spanish Label Timer System (Player-specific)**

- Added displayTimer parameter (3 seconds default)
- Added cooldownTimer parameter (5 seconds default)

- Timer auto-hides player label after X seconds
- Cooldown prevents spam (label won't show again for X seconds)
- Configurable per object (0 = disabled, stays visible)
- Perfect for player UX (brief info, not intrusive)

Timer Implementation Details:

- **displayTimer:** Show label for X seconds, then auto-hide
- **cooldownTimer:** Wait X seconds before showing again
- Works with both proximity detection AND triggers
- **Player:** displayTimer = 3s (shows briefly)
- **Enemy:** displayTimer = 0s (always visible when near)
- **Tilemap objects:** displayTimer = 0s (visible in trigger zone)

Bug Fixes:

- Fixed coroutine error when panel inactive
- Added safety check: `!labelPanel.activeSelf`
- Fixed screen position warning (cosmetic)
- Resolved player damage collision issue

Unity Implementation Completed:

- ☒ UI Panel created with Best Education B.V. logo
- ☒ Logo Image component (not text) - professional look
- ☒ SpanishObjectLabel added to Player (with timer)
- ☒ SpanishObjectLabel added to Enemy prefab
- ☒ Invisible triggers created for tilemap objects
- ☒ All scripts tested and working
- ☒ No critical bugs remaining

Educational System Verification:

- ☒ 20+ Spanish vocabulary words available
- ☒ Logo branding on all labels
- ☒ Smooth fade animations
- ☒ Timer prevents label spam for player
- ☒ System fully functional and polished

MoSCoW Final Status:

- **MUSTS:** 6/6 - 100% ☒
- **SHOULDs:** 2/2 - 100% ☒
- **COULDs:** 0/2 - Niet vereist
- **WOULDs:** 0/1 - Niet vereist

REFLECTIE EINDPROJECT

Dit project heeft een complete transformatie ondergaan:

- **Week 1-3:** Basic gameplay mechanics (movement, combat, stats)
- **Week 4:** Enemy AI state machine (game-changer)
- **Week 5:** Spanish learning system + branding (unique differentiator)
- **Week 6:** Timer polish + final bug fixes

Wat maakt deze game uniek:

1. **Educational value** - learn Spanish through gameplay
2. **Best Education B.V. branding** throughout
3. **Smart timer system** - non-intrusive UX
4. **Professional polish** - smooth animations, proper feedback
5. **Scalable architecture** - easy to add more vocabulary

Technical achievements:

-  State machine AI (Patrol/Chase/Attack/Idle)
-  Dynamic difficulty scaling (enemy stats per player level)
-  Dual detection system (triggers + proximity)
-  Timer/cooldown system for optimal UX
-  Scene persistence (DontDestroyOnLoad)
-  Complete UI/UX with branding
-  Educational integration

EINDCIJFERS

Total Development Time: ~6 weken (30 aug - 8 okt) **Total Commits:** ~70+